



Supplemental S1 — Literature Review (auto-composed from deep search)

*Composed by `scripts/s_compose_literature_review.py` from the most recent deep-search run. Edit the script, not this file — manual edits will be overwritten on the next pipeline run.*

This review covers **3 keyword(s)**, **300 unique paper(s)** (retrieved at up to 100 per keyword from arxiv, crossref). All references are stored in `manuscript/references_deep.bib` and resolved by the combined-PDF pipeline (Pandoc `--natbib` + BibTeX over all `manuscript/*.bib` files).

*Papers retrieved: **100** · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*



# convex optimization II

| Cite                               | Title                                                   | Year DOI / URL                                    |
|------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| (Cambridge University Press 2004c) | Convex optimization problems                            | 2004 DOI 10.1017/cbo9780511804441.001 (open)      |
| (Cambridge University Press 2004b) | Convex functions                                        | 2004 DOI 10.1017/cbo9780511804441.002 (open)      |
| (Springer-Verlag, n.d.-e)          | Reverse Convex Optimization, Reverse convex programming | n.d. DOI 10.1007/springerreference_72642 (open)   |
| (Suh 2022)                         | 1. Convex Optimization Basics                           | 2022 DOI 10.1561/9781638280538.ch1 (open)         |
| (Cambridge University Press 2021c) | Convex Optimization and Efficiency                      | 2021 DOI 10.1017/9781108699211.006 (open)         |
| (Cambridge University Press 2021d) | Ellipsoid Method for Convex Optimization                | 2021 DOI 10.1017/9781108699211.015 (open)         |
| (Nesterov 2004b)                   | Smooth Convex Optimization                              | 2004 DOI 10.1007/978-1-4419-8853-9_2 (open)       |
| (Nesterov 2004a)                   | Nonsmooth Convex Optimization                           | 2004 DOI 10.1007/978-1-4419-8853-9_3 (open)       |
| (Emerald Publishing Limited 2022)  | Convex Optimization Basics                              | 2022 DOI 10.1561/978-1-63828-053-820251002 (open) |
| (Cambridge University Press 2004d) | Convex sets                                             | 2004 DOI 10.1017/cbo9780511804441.003 (open)      |
| (CRC Press 2011g)                  | What Is Convex Optimization?                            | 2011 DOI 10.1201/b11156-7 (open)                  |
| (CRC Press 2011c)                  | Convex Semi-Infinite Optimization                       | 2011 DOI 10.1201/b11156-17 (open)                 |
| (CRC Press 2011d)                  | Tools for Convex Optimization                           | 2011 DOI 10.1201/b11156-8 (open)                  |

*Papers retrieved: **100** · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*





# stochastic gradient descent II

| Cite                                | Title                                                                                             | Year DOI / URL                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------|
| (Gwózdź 2018)                       | Stochastic and semi-stochastic gradient descent in speech disambiguation                          | 2018 DOI 10.7490/f1000research.1115776 (open)      |
| (Wang and Murphy 2018)              | Stochastic gradient descent                                                                       | 2018 DOI 10.53347/rid-61715 (open)                 |
| (Kosinski 2017)                     | coxphSGD: Stochastic Gradient Descent log-Likelihood Estimation in Cox Proportional Hazards Model | 2017 DOI 10.32614/cran.package.coxphsgd (open)     |
| (PeerJ, n.d.-d)                     | Table 4: Stochastic gradient descent (SGD) parameter settings with CC images.                     | n.d. DOI 10.7717/peerj-cs.3332/table-4 (open)      |
| (Peseux et al. 2023)                | Stochastic Gradient Descent with Gradient Estimator for Categorical Features                      | 2023 DOI 10.2139/ssrn.4439301 (open)               |
| (Theodoridis 2015)                  | Stochastic Gradient Descent                                                                       | 2015 DOI 10.1016/b978-0-12-801522-3.00005-7 (open) |
| (Cambridge University Press 2014)   | Stochastic Gradient Descent                                                                       | 2014 DOI 10.1017/cbo9781107298019.01 (open)        |
| (SPIE-Intl Soc Optical Eng, n.d.-a) | 10.1117/12.2305101.5783306294001                                                                  | n.d. DOI 10.1117/12.2305101.578330629 (open)       |
| (Lan 2023)                          | Stochastic Gradient Descent                                                                       | 2023 DOI 10.1007/978-3-030-54621-2_777-1 (open)    |
| (SPIE-Intl Soc Optical Eng, n.d.-b) | 10.1117/12.2512817.6013939792001                                                                  | n.d. DOI 10.1117/12.2512817.601393979 (open)       |
| (Gong 2026)                         | A Comparative Study of Gradient Descent and Stochastic Gradient Descent in                        | 2026 DOI 10.54254/2755-2721/2026.Id35719 (open)    |

*Papers retrieved: **100** · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*



# reproducible research II

| Cite                                   | Title                                                      | Year DOI / URL                               |
|----------------------------------------|------------------------------------------------------------|----------------------------------------------|
| (Gandrud 2018c)                        | Introducing Reproducible Research                          | 2018 DOI 10.1201/9781315382548-1 (open)      |
| (Gandrud 2020c)                        | Introducing Reproducible Research                          | 2020 DOI 10.1201/9780429031854-2 (open)      |
| (Gandrud 2018b)                        | Getting Started with Reproducible Research                 | 2018 DOI 10.1201/9781315382548-2 (open)      |
| (Gandrud 2020b)                        | Getting Started with Reproducible Research                 | 2020 DOI 10.1201/9780429031854-3 (open)      |
| (Hoefling and Rossini 2018)            | Reproducible Research for Large-Scale Data Analysis        | 2018 DOI 10.1201/9781315373461-8 (open)      |
| (Xie 2018)                             | knitr: A Comprehensive Tool for Reproducible Research in R | 2018 DOI 10.1201/9781315373461-1 (open)      |
| (Baker 2022)                           | Reproducible data analysis                                 | 2022 DOI 10.1093/hesc/9780192896599.0 (open) |
| (PubPub, n.d.)                         | NinBioinformatics                                          | n.d. DOI 10.21428/3c290898 (open)            |
| (University of California Press 2019d) | Reproducible Research Reports                              | 2019 DOI 10.2307/j.ctvpb3xkg.16 (open)       |
| (Preeyanon et al. 2018)                | Reproducible Bioinformatics Research for Biologists        | 2018 DOI 10.1201/9781315373461-7 (open)      |
| (University of California Press 2019a) | ELEVEN. Reproducible Workflow                              | 2019 DOI 10.1525/9780520969230-014 (open)    |
| (Hrynaskiewicz et al. 2018)            | Open Science and the Role of                               | 2018 DOI                                     |

*Per-paper synthesis omitted — no LLM Contribution paragraphs are present in the deep-search outputs (set `project_config.deep_search.llm_per_paper: true` and ensure Ollama is reachable to populate this section).*

*Composition summary: 3 keywords · 300 unique papers · 300 per-paper notes integrated · 300 BibTeX entries · 0 key(s) missing from bib.*